

ECEN 714

Lab1: Introduction to Cadence Schematic Design & Simulation

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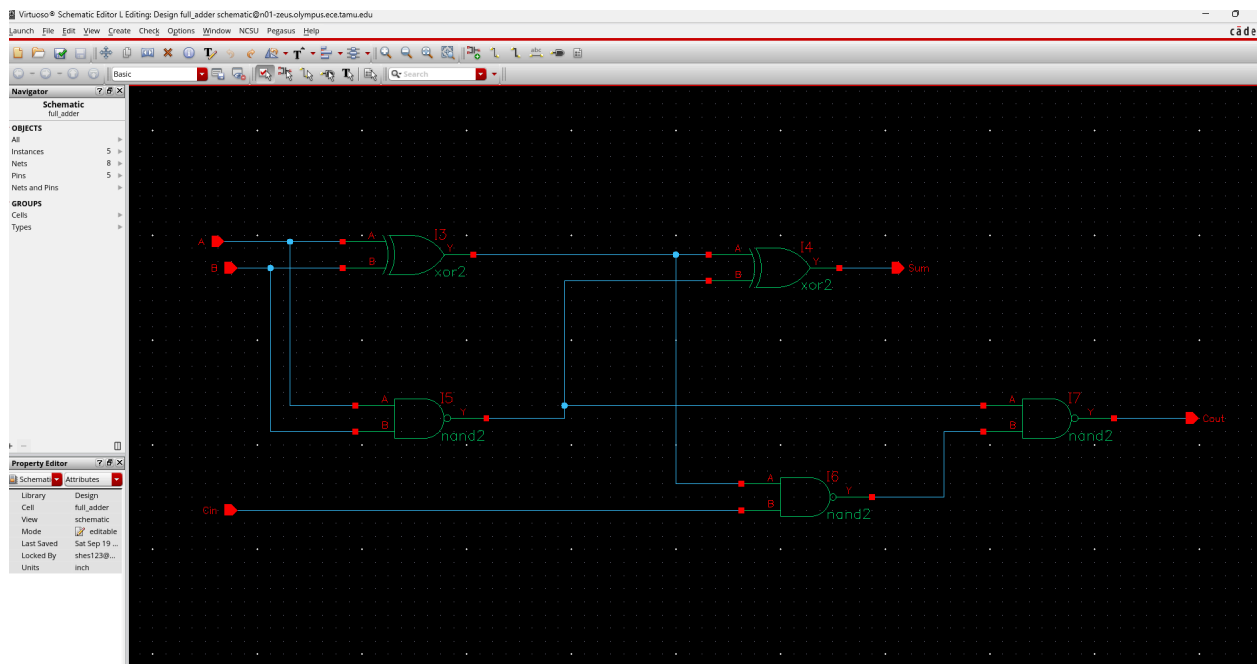
UIN:537000654

TA:Lilith Xu

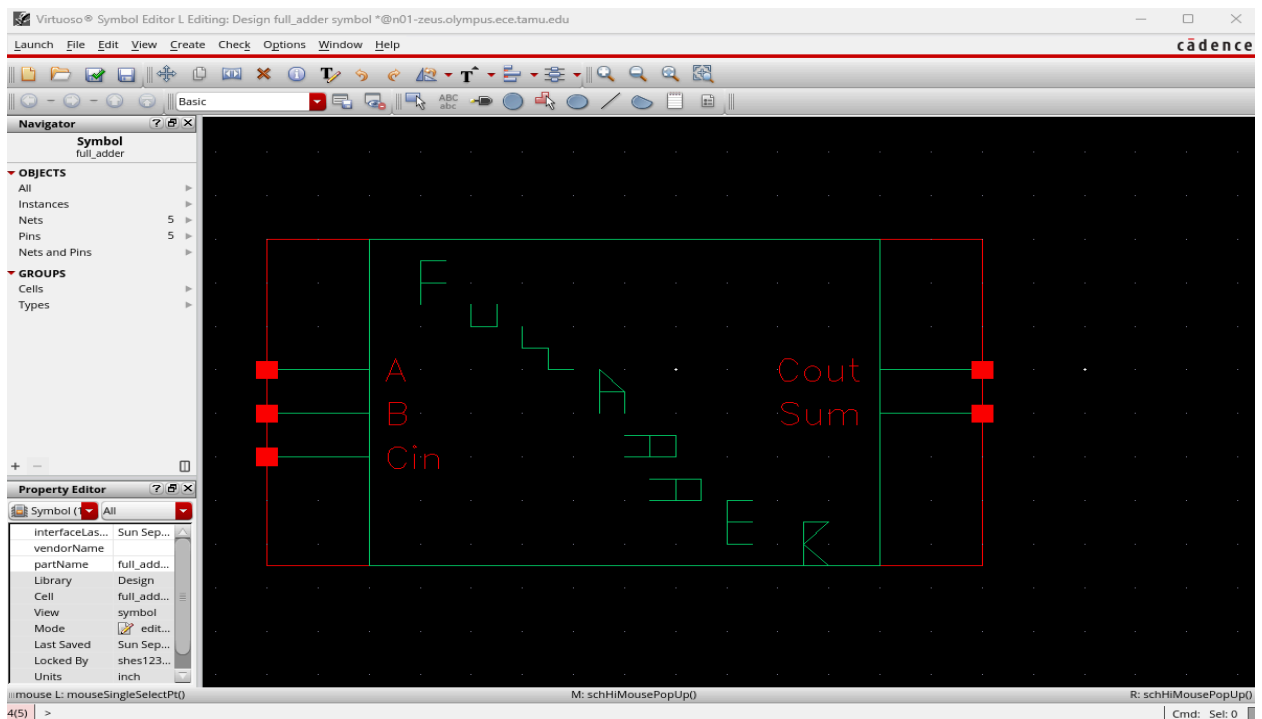
Lab section:609

A. Full Adder

- Schematic:



- Symbol:

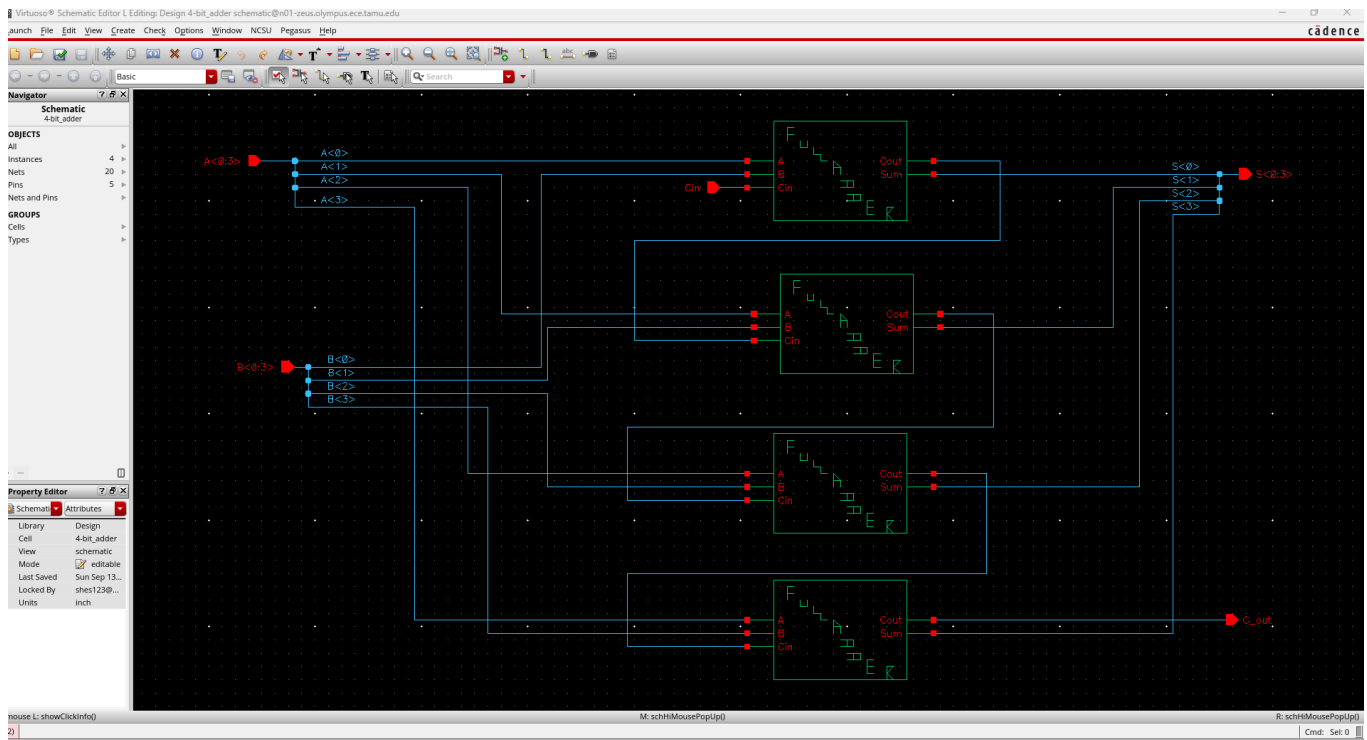


- Result and Explanation:

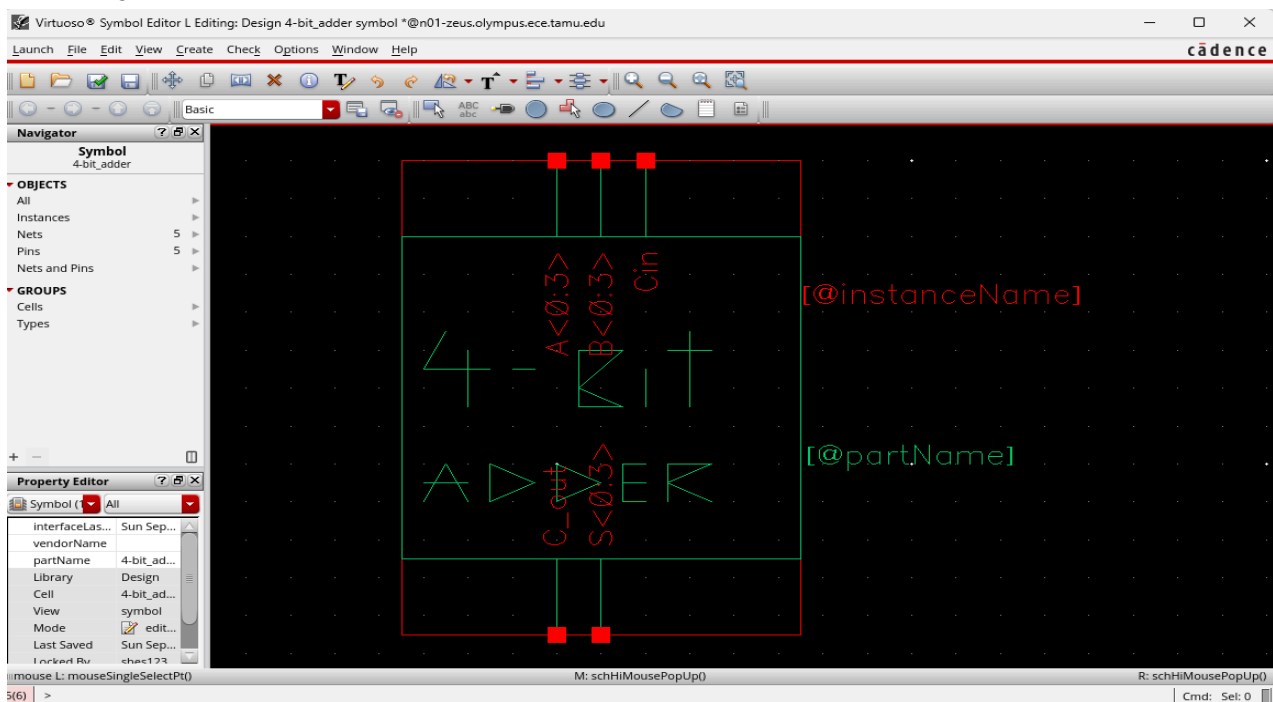
Created Basic full adder with nand gates and xor gates alone

B. 4-bit Full Adder

- Schematic:



- **Symbol:**

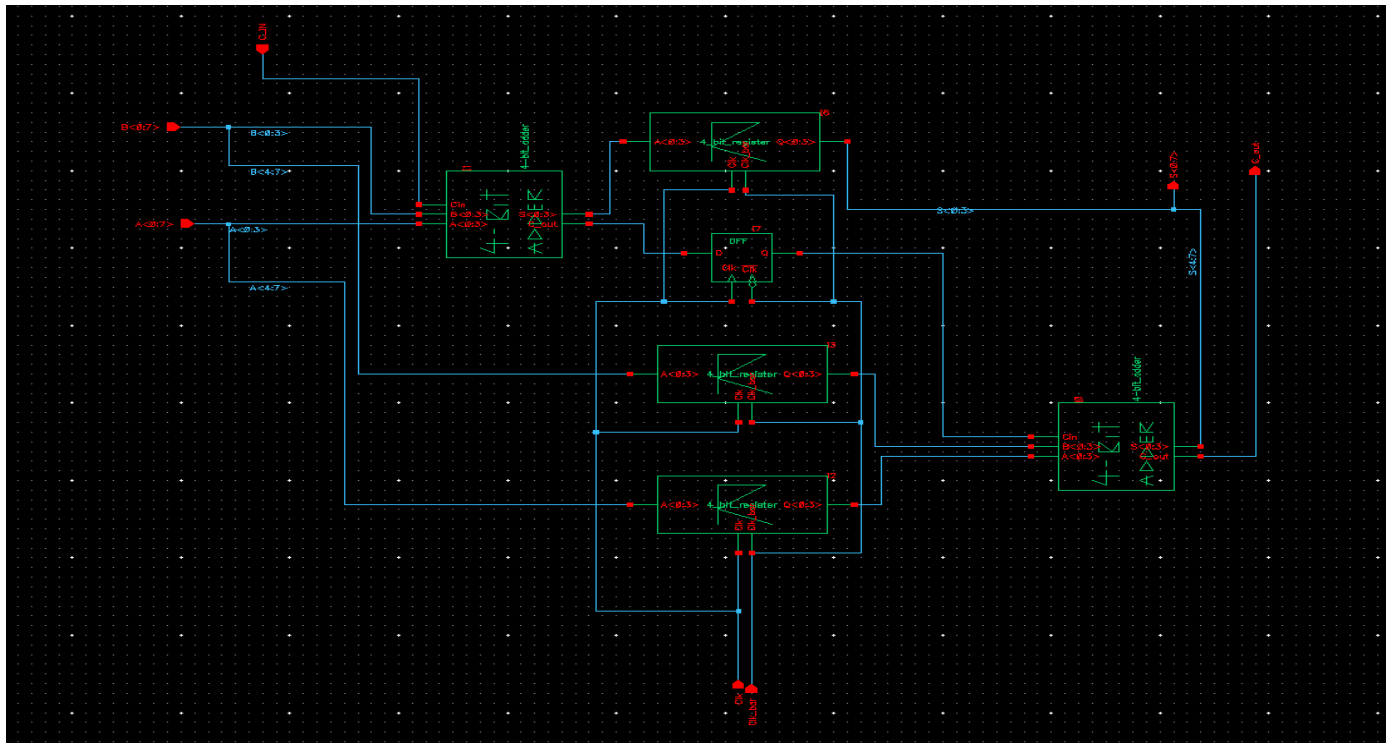


- **Result and Explanation:**

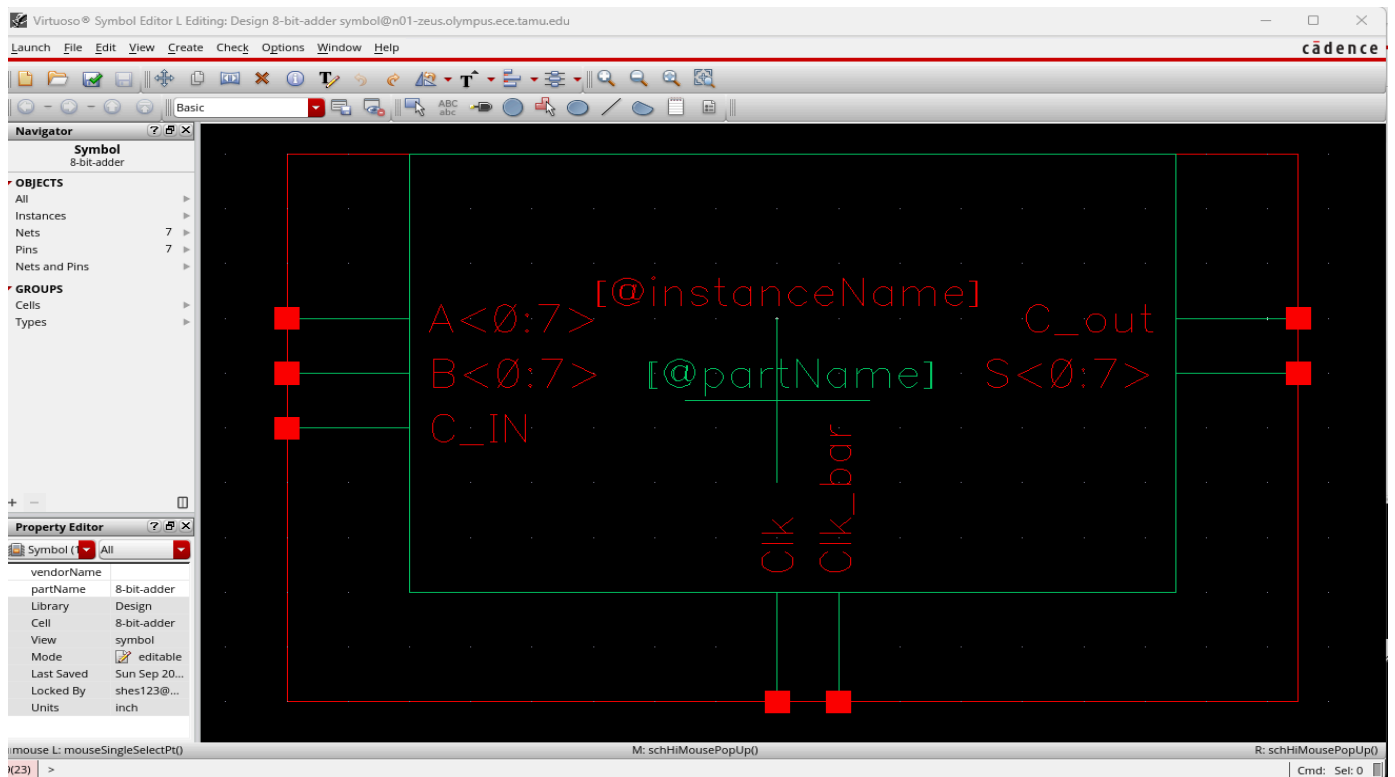
Using the full adder module ,4-bit adder is been designed

C. 8-bit Pipelined Adder:

- **Schematic:**



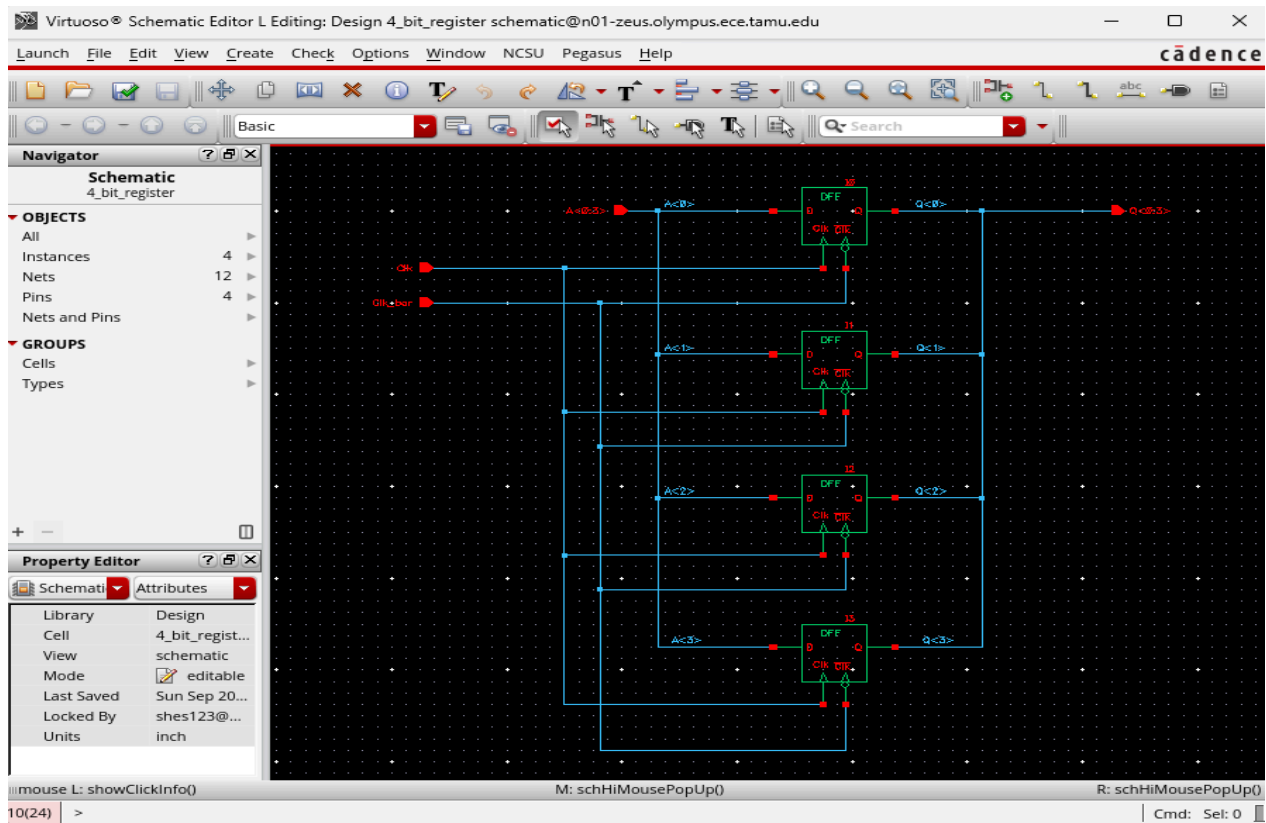
- **Symbol:**



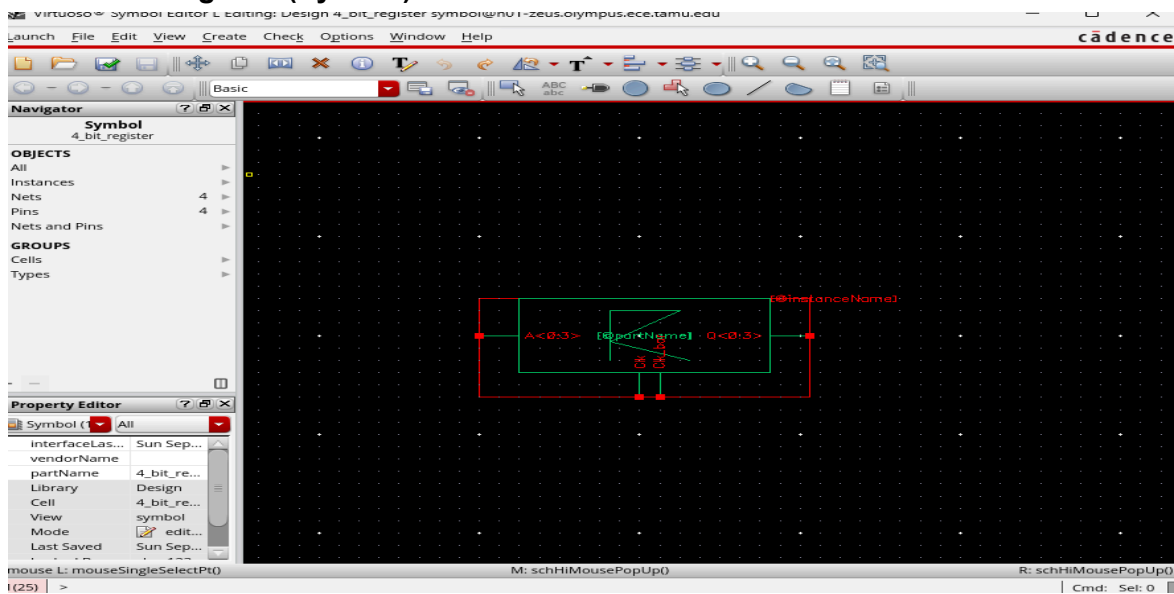
- **Result and Explanation:**

Using the basic blocks of 4-bit adder, its been pipelined with pipeline registers to form 2 stage kind of 8 bit adder with 2 blocks of 4-bit adder

- Design of pipelining Registers:
- 4-bit register:(Schematic)



- 4-bit register:(Symbol):



Conclusion:Have done the block level design and gate level designof the above lab